

Computer Security Capstone

Project II: MITM and Pharming Attacks in Wi-Fi Networks

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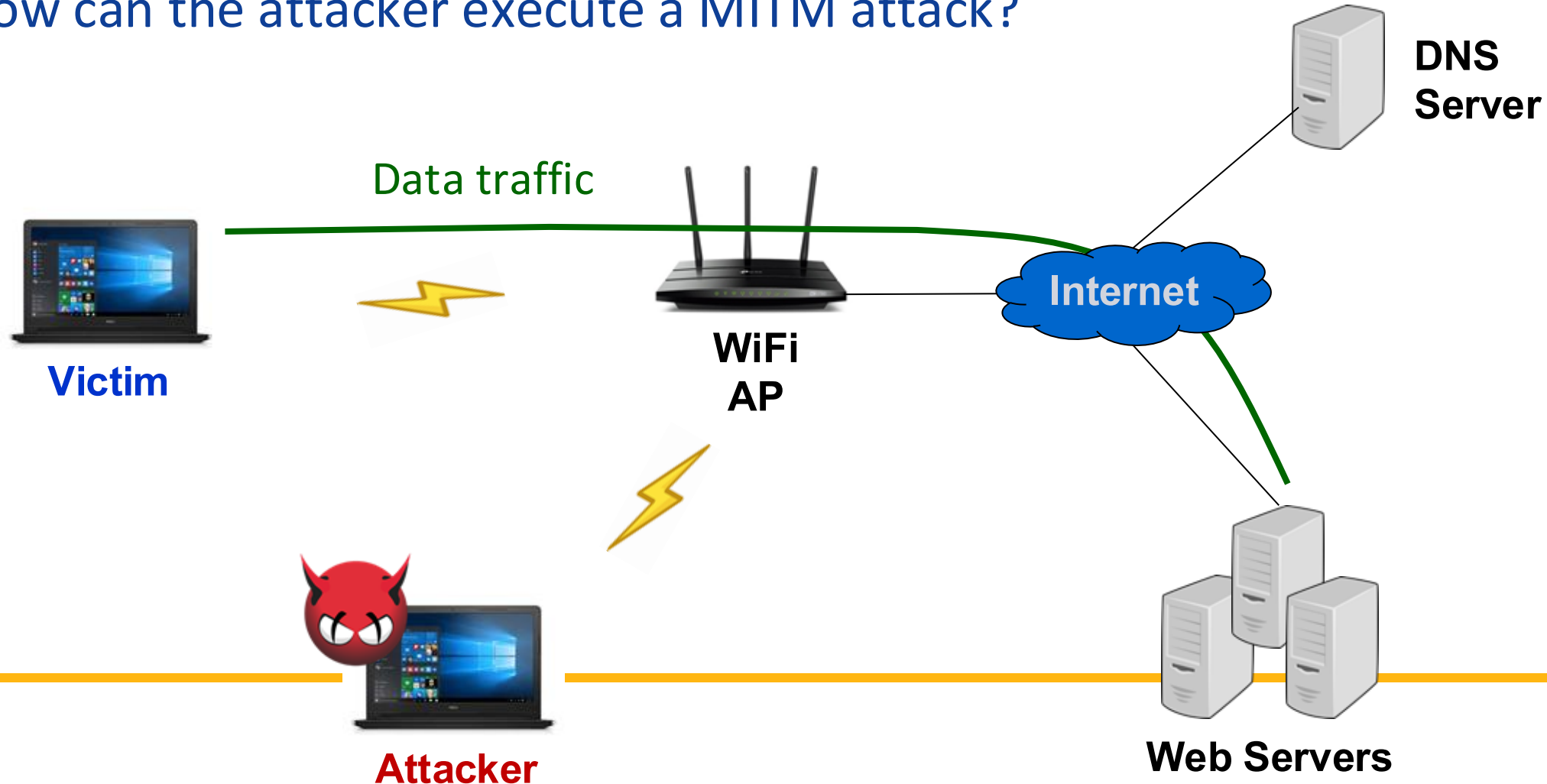
Goal

- Understand how users can be targeted by a man-in-the-middle (MITM) attack over Wi-Fi networks
- You will learn how to
 - ☐ scan IP/MAC addresses of the devices in a Wi-Fi network
 - ☐ launch a man-in-the-middle (MITM) attack by using ICMP redirect
 - ☐ launch a pharming attack



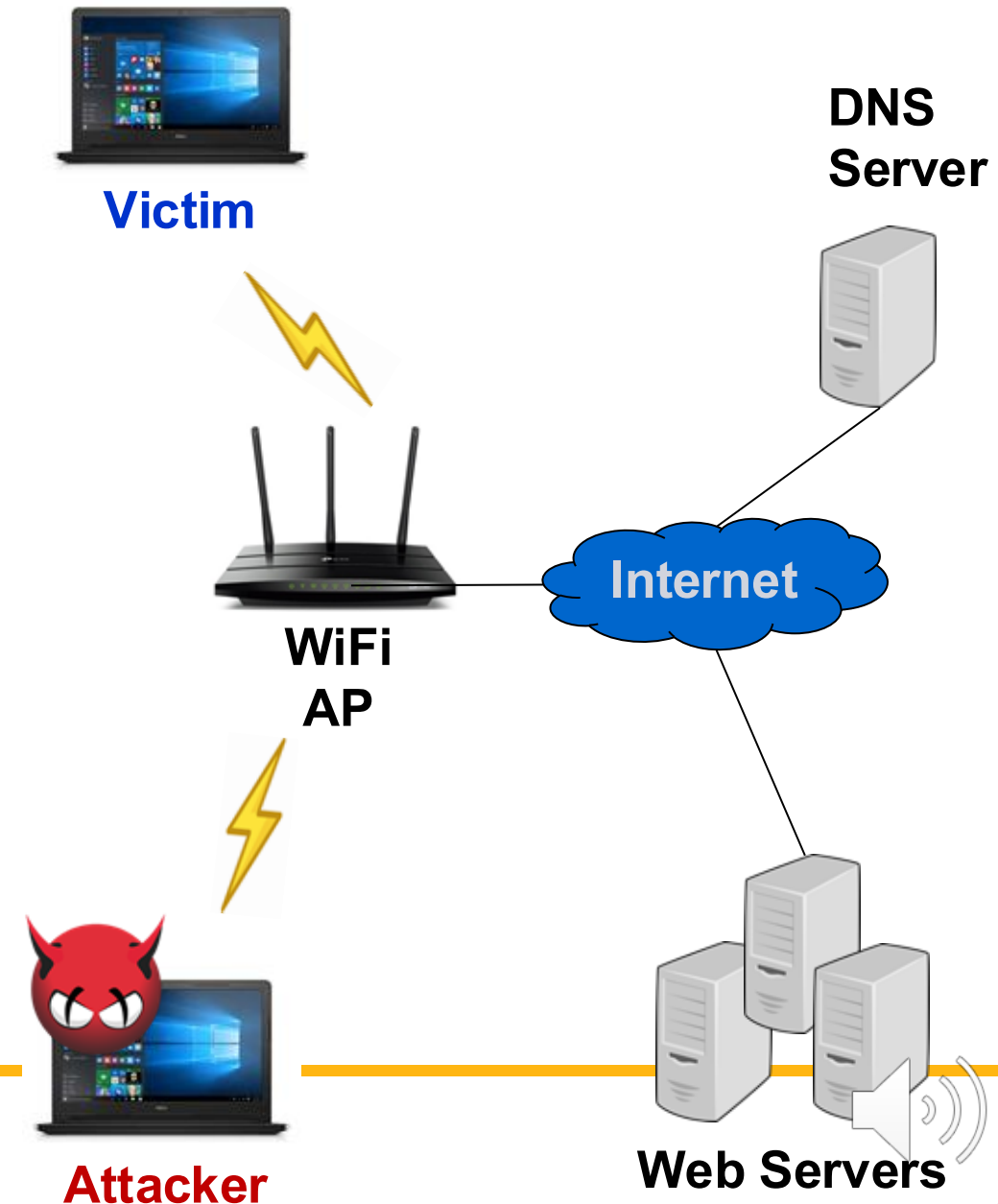
Attack Scenario

- How can the attacker execute a MITM attack?



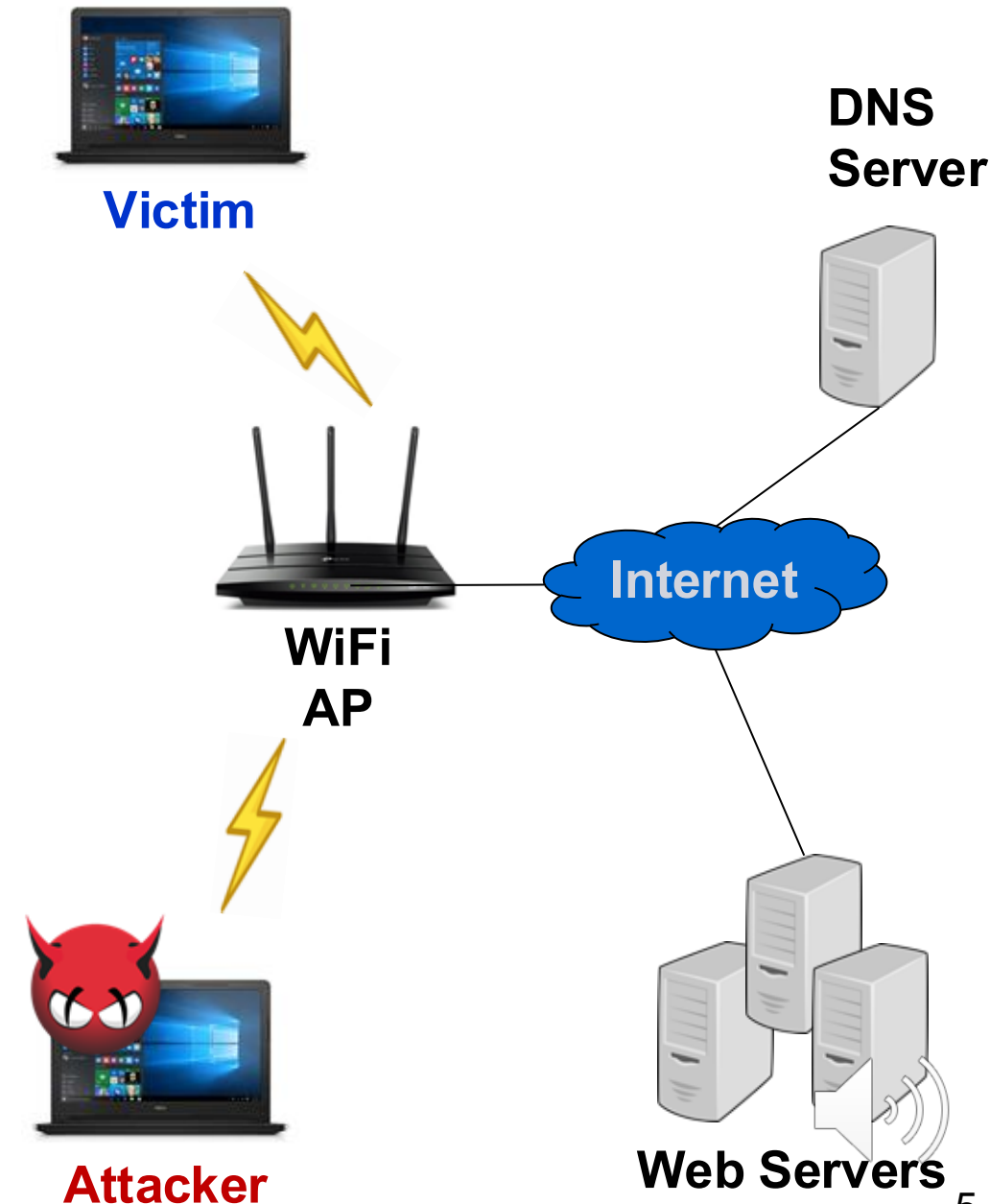
Major Ideas

- Identify the Victim
 - ☐ Device Address Information Collection
- Redirect Victim Traffic
 - ☐ ICMP redirect attack
- Launch Pharming Attack
 - ☐ DNS Spoofing



Tasks: MITM and Pharming

- Device Address Information Collection (Task I: 10%)
 - ☐ Obtain all other client devices' IP/MAC addresses in a connected Wi-Fi network
- ICMP redirect attack (Task II: 10%)
 - ☐ ICMP redirect for specific client device in the Wi-Fi network
- DNS Spoofing (Task III: 15%+Task IV 15%)
 - ☐ DNS spoofing attack for web services
 - ☐ Bypass DoH protection
- Some implementation questions during the demo (50%)



Task I: Device Address Information Collection

- Scan all the devices' IP/MAC addresses in the Wi-Fi network
 - 🔍 Hint: ARP format and raw socket.
- Fetch the IP/MAC addresses of all the other client devices

```
● csc2025-attacker@csc2025-attacker:~/Desktop/csc2025-project2$ sudo ./icmp_redirect 163.182.194.25
Available devices
-----
Index | IP      | MAC
-----
0     | 10.0.2.1 | 52:54:00:12:35:00
1     | 10.0.2.2 | 52:54:00:12:35:00
2     | 10.0.2.3 | 80:00:27:36:DA:66
3     | 10.0.2.16 | 80:00:27:6F:58:50
-----
```



Task II: ICMP redirect attack

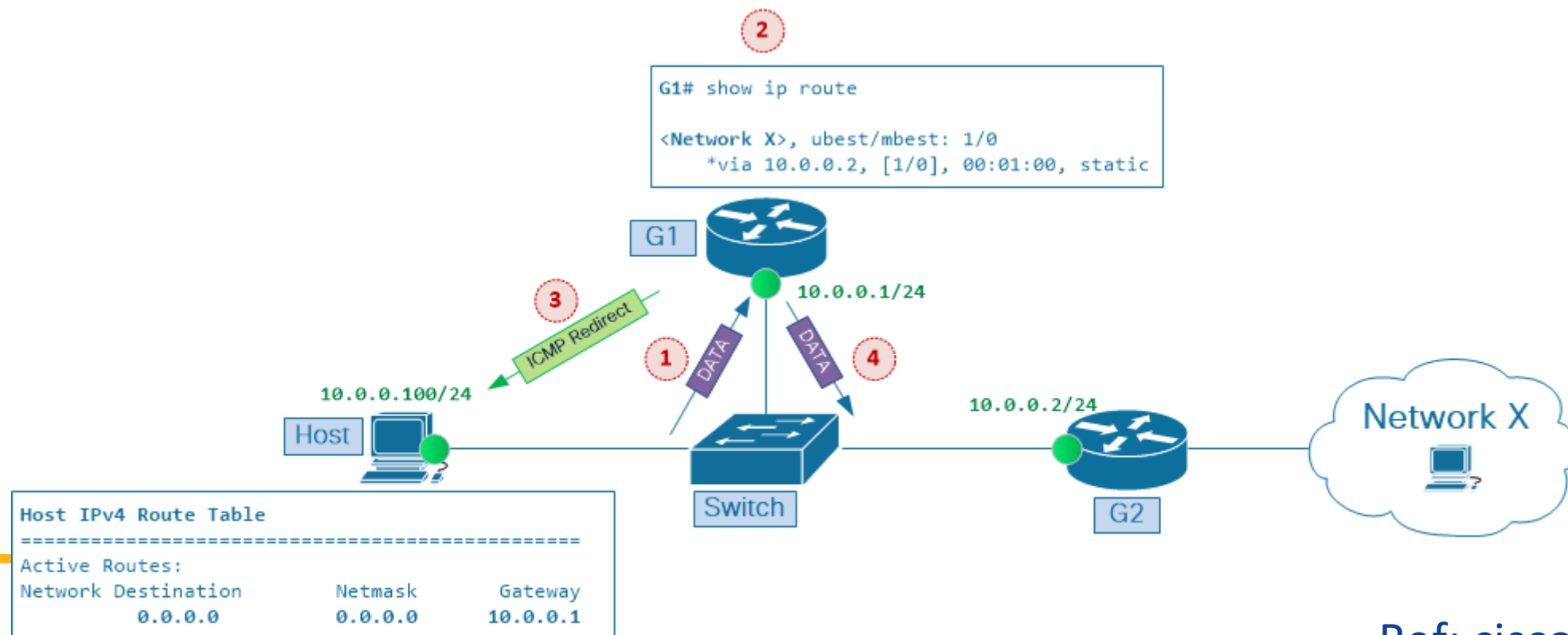
- What is ICMP redirect?

- It is a type of ICMP (Internet Control Message Protocol) message used to inform a host to update its routing information.
 - Purpose: Informs the host of a better route for its traffic.
 - Trigger Condition: Occurs when a router detects unnecessary routing (e.g., packets looping back to the same network segment).
 - Message Content: Contains the IP address of the new next-hop router.
 - Common Use: Optimizes routing by reducing unnecessary hops.



Task II: ICMP redirect attack (Cont.)

- Router G1 identifies a better route through the G2 and sends an ICMP redirect to the host to update its routing table.

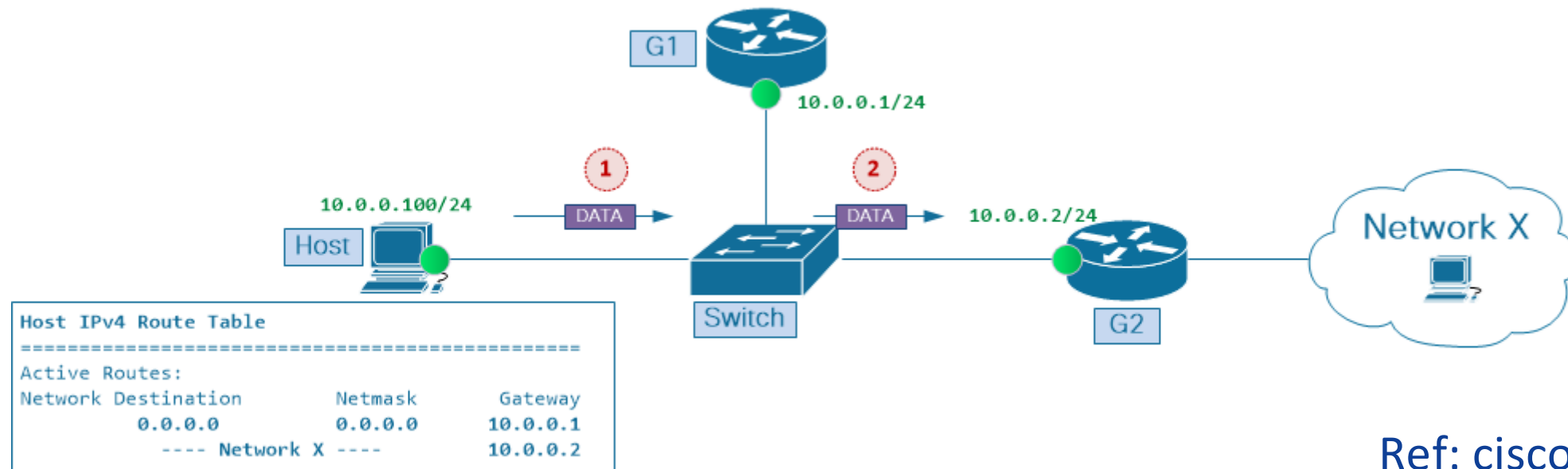


Ref: [cisco](#)



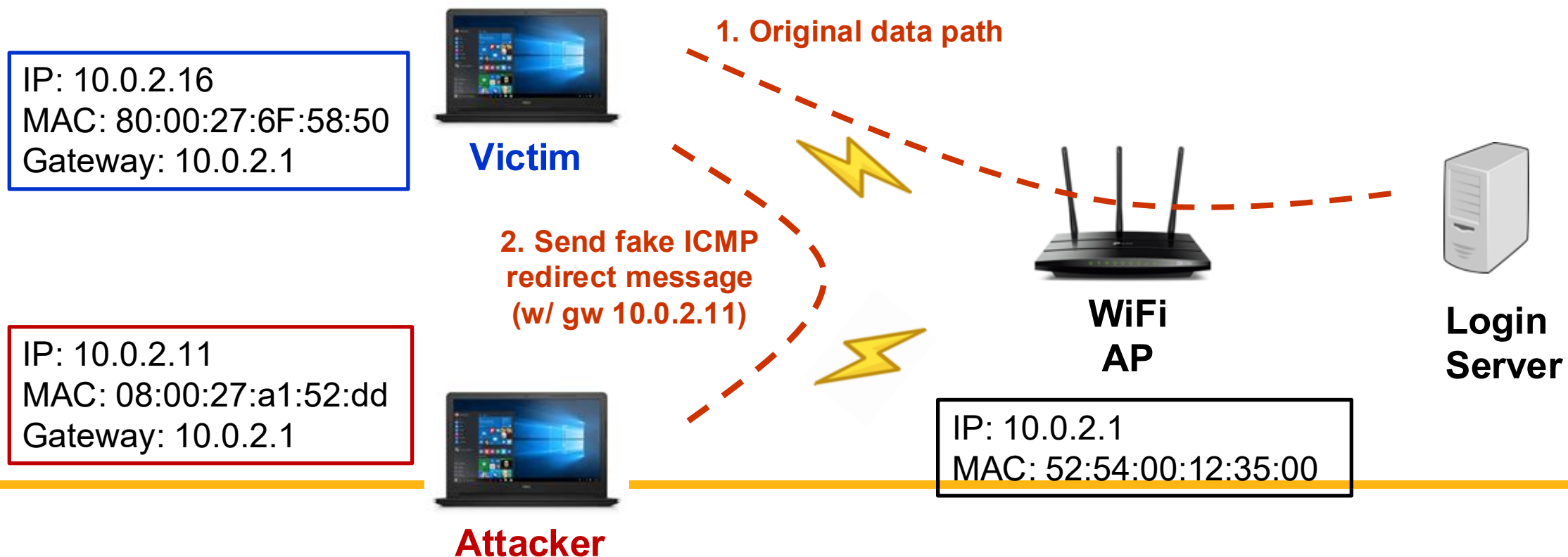
Task II: ICMP redirect attack (Cont.)

- The host updates its routing table and sets the gateway to G2.



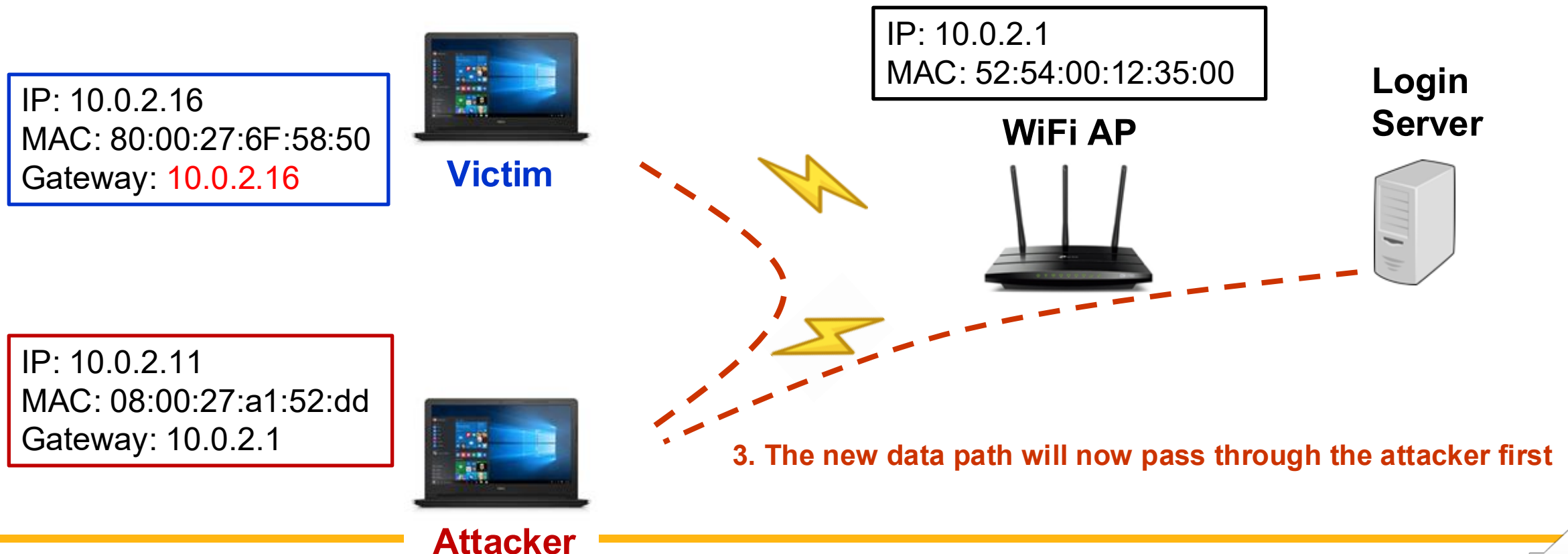
Task II: ICMP redirect attack (Cont.)

- The attacker generates a fake ICMP redirect packet, claiming there is a better route for the victim's traffic.



Task II: ICMP redirect attack (Cont.)

- The victim updates its routing table and sets the gateway as the attacker.



Task II: ICMP redirect attack (Cont.)

- Attacker scans devices and sends fake ICMP redirect message to the victim.

```
csc2025-attacker@csc2025-attacker:~/Desktop/csc2025-project2$ sudo ./icmp_redirect 163.182.194.25
Available devices
-----
Index | IP | MAC
-----
0 | 10.0.2.1 | 52:54:00:12:35:00
1 | 10.0.2.2 | 52:54:00:12:35:00
2 | 10.0.2.3 | 80:00:27:36:DA:66
3 | 10.0.2.16 | 80:00:27:6F:58:50
-----
Select Victim IP index: 3
Select Gateway IP index: 0
Victim IP: 10.0.2.16, Gateway IP: 10.0.2.1, Attacker IP: 10.0.2.11
ICMP Redirect packet sent successfully!
```

- The victim updates its routing table and sets the gateway as the attacker.

```
csc2025-victim@csc2025-victim:~$ ip route get 163.182.194.25
163.182.194.25 via 10.0.2.11 dev enp0s3 src 10.0.2.16 uid 1000
cache <redirected> expires 296sec
```

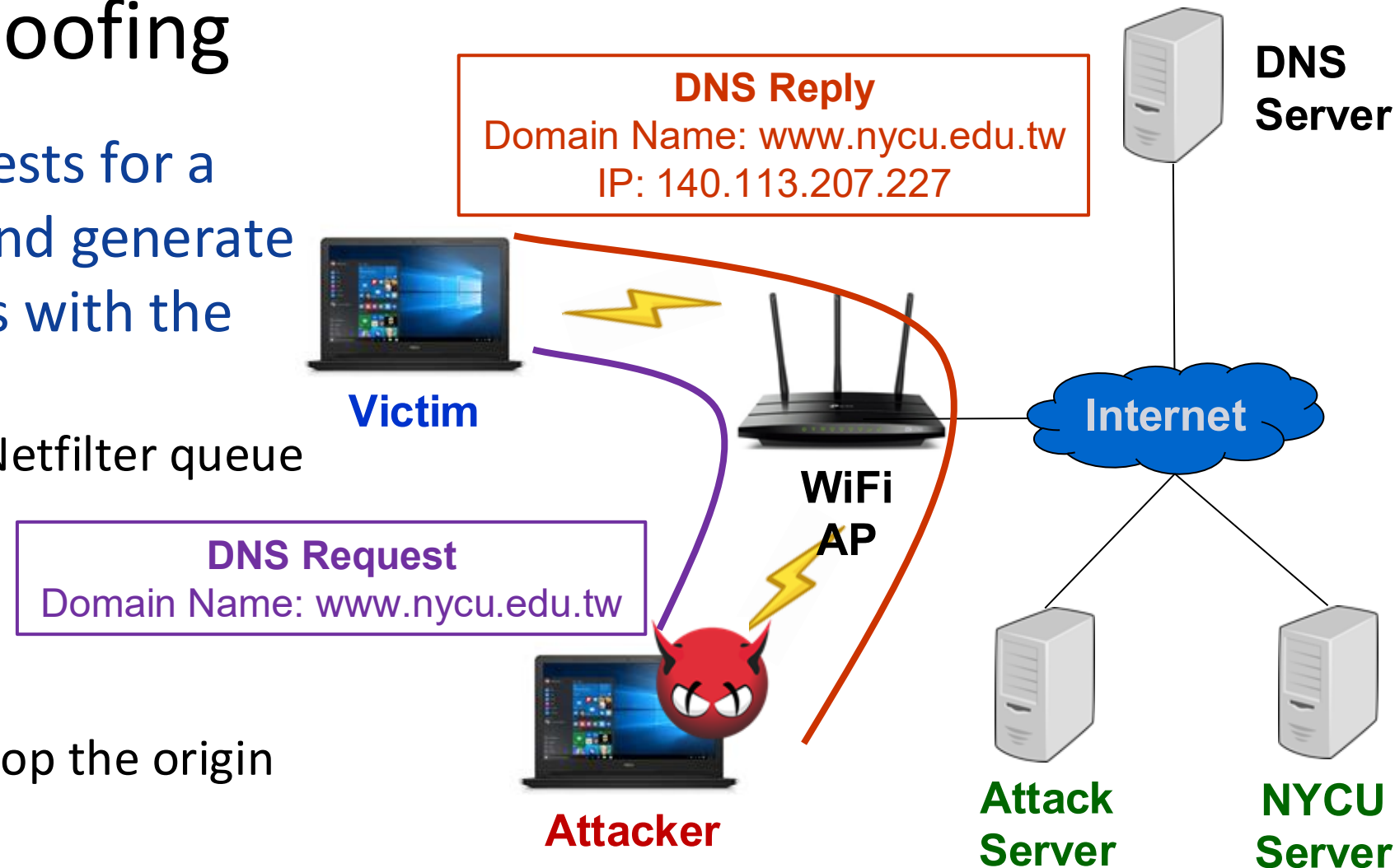


Task III: DNS Spoofing

- Intercept DNS requests for a specific web page and generate spoofed DNS replies with the attack server's IP

❓ Hint: DNS format, Netfilter queue

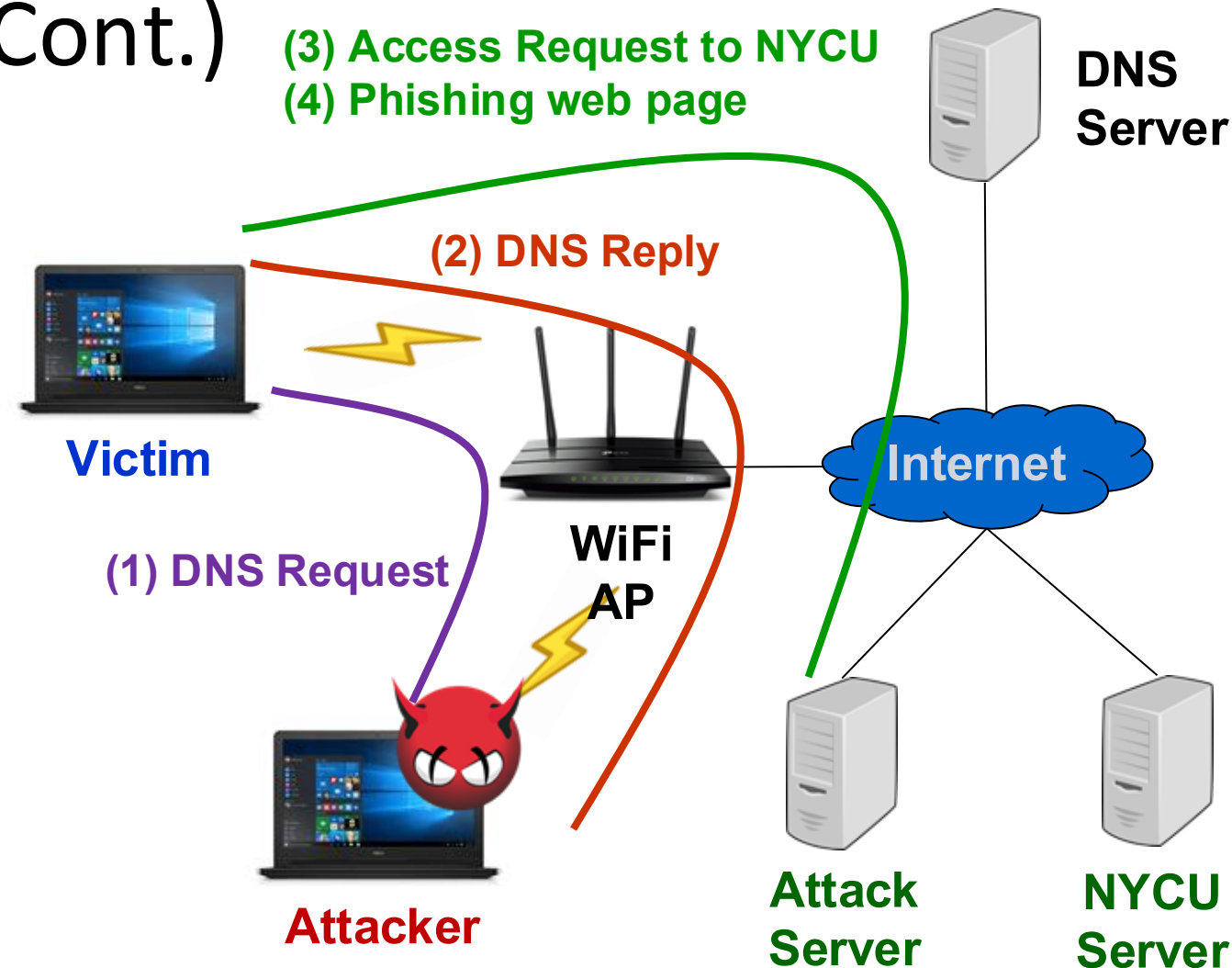
❓ Note: You should drop the origin victim DNS request



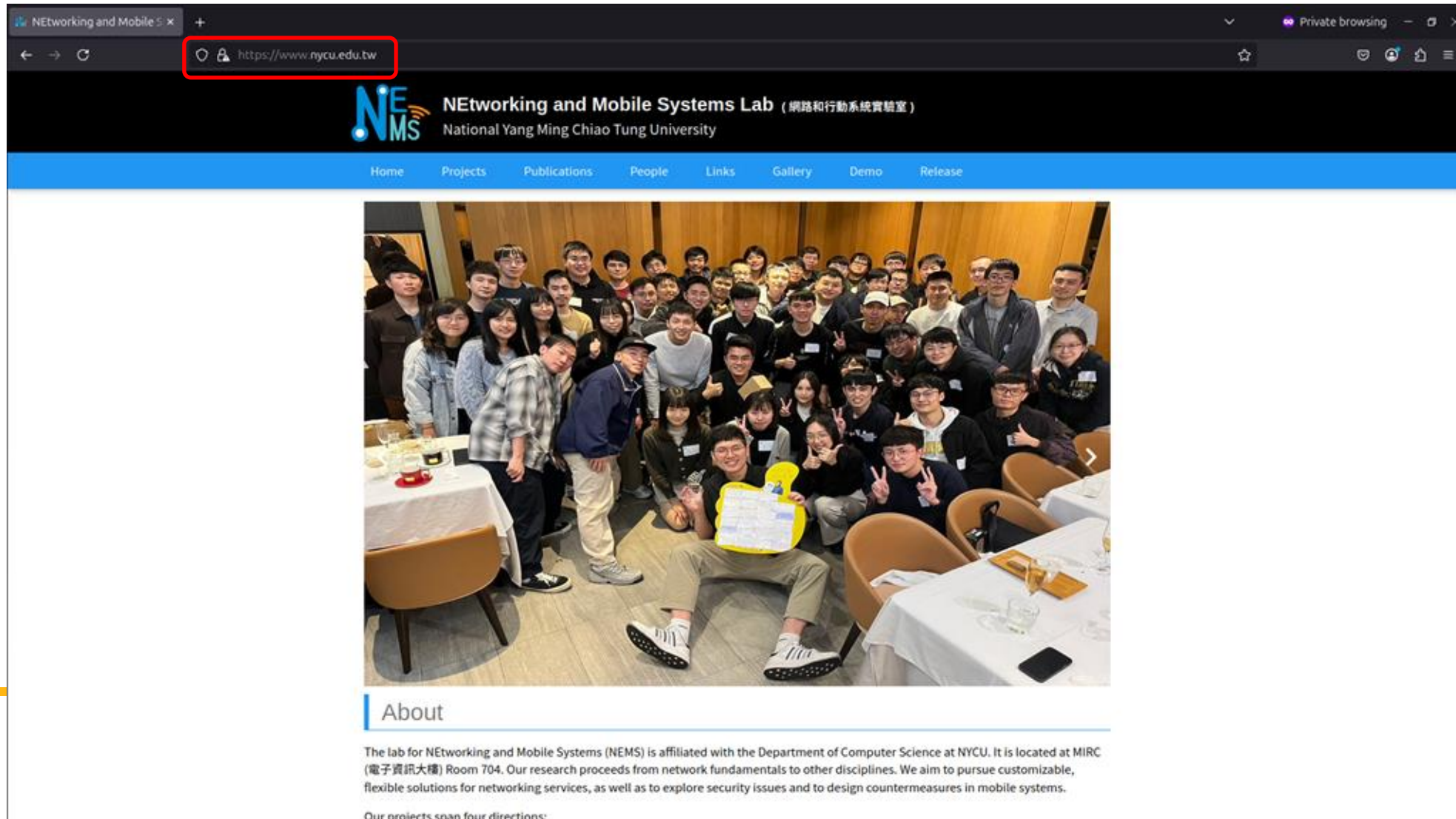
Task III: DNS Spoofing (Cont.)

• Successful attack

- ❑ An access request to NYCU home page will be redirected to the attack server
- ❑ A phishing web page will be shown to Victim
 - please redirect the victim to our lab's website (140.113.207.227)



Task III: DNS Spoofing (Cont.)



Networking and Mobile Systems Lab (網路和行動系統實驗室)
National Yang Ming Chiao Tung University

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About

The lab for NETworking and Mobile Systems (NEMS) is affiliated with the Department of Computer Science at NYCU. It is located at MIRC (電子資訊大樓) Room 704. Our research proceeds from network fundamentals to other disciplines. We aim to pursue customizable, flexible solutions for networking services, as well as to explore security issues and to design countermeasures in mobile systems.

Our projects span four directions:

Task IV: DoH Spoofing

- What is DNS over HTTPS (DoH)?

- ❑ Encrypts DNS queries inside HTTPS to prevent eavesdropping
- ❑ Bypasses traditional DNS interception (UDP port 53)
- ❑ Used by default in modern browsers (Firefox, Chrome)

- Why does it break Task III?

- ❑ Task III intercepts UDP port 53 — DoH never uses UDP 53
- ❑ Firefox sends DNS queries directly to Cloudflare via HTTPS
- ❑ The attacker must intercept HTTPS traffic instead



Task IV: DoH Spoofing (Cont.)

Traditional DNS



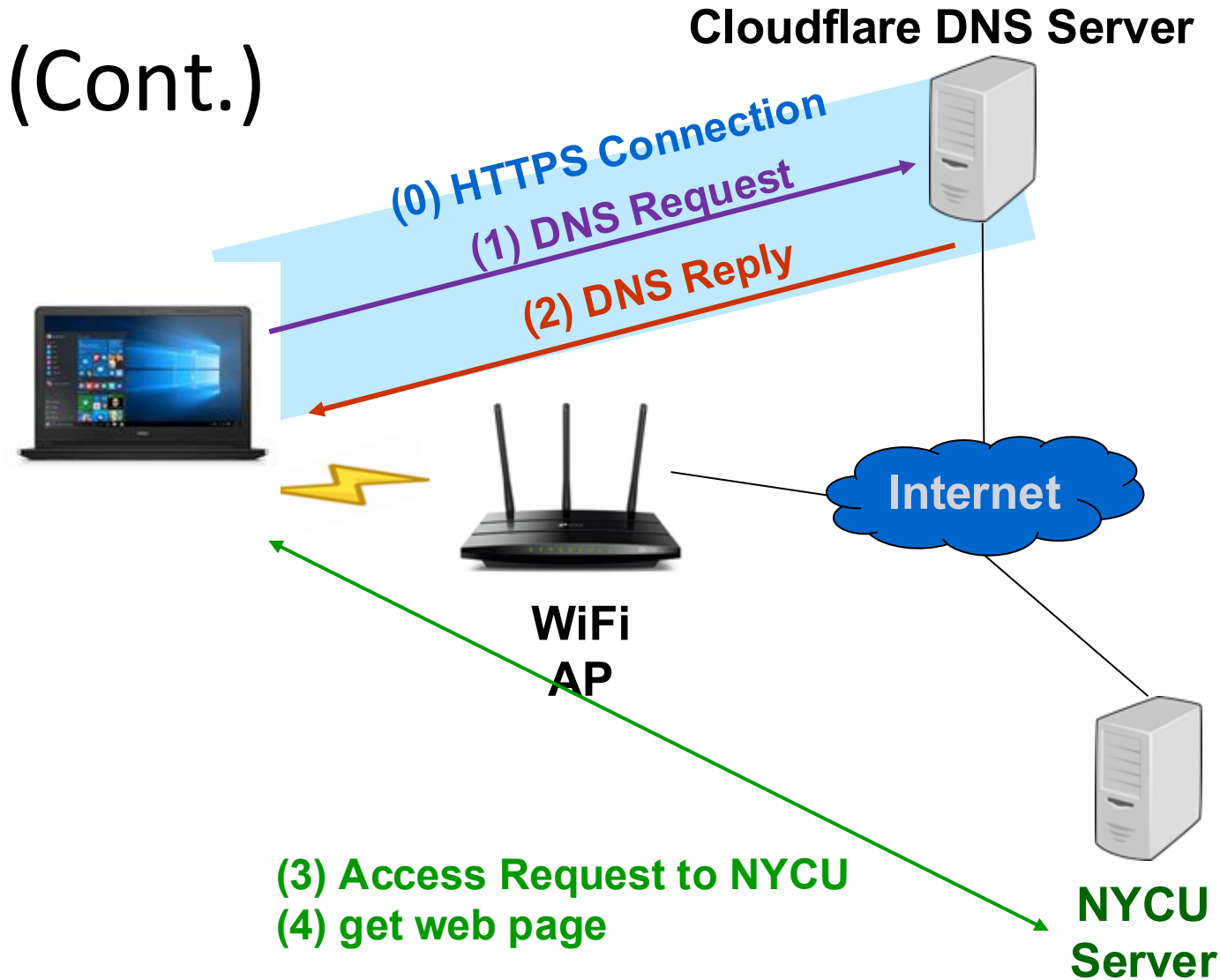
DoH, DNS over HTTPS



Task IV: DoH Spoofing (Cont.)

• Normal Scenario

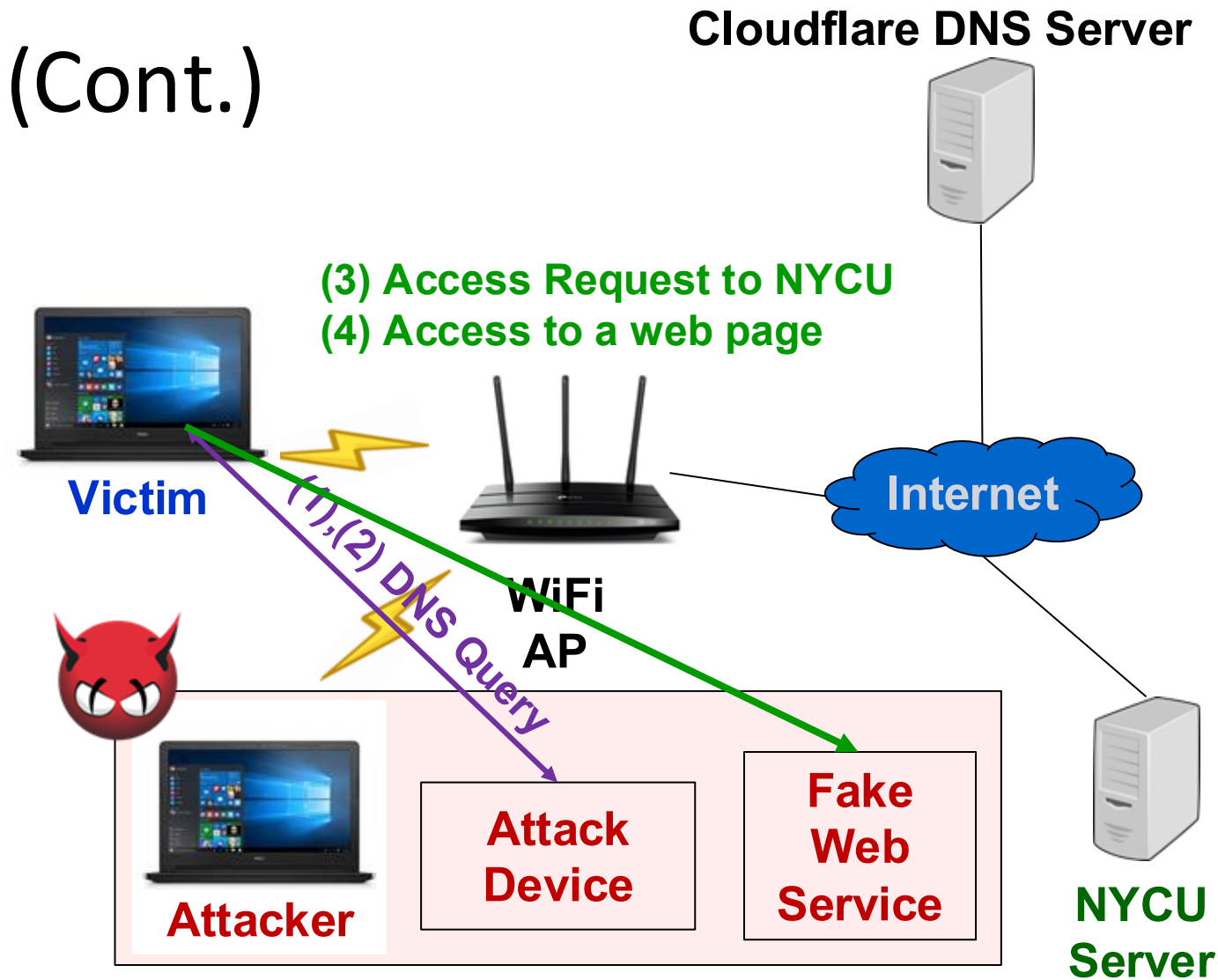
- ❑ The browser establishes an HTTPS connection to Cloudflare's DoH Server.
- ❑ DNS queries are encrypted!
- ❑ Although you are in the middle of the path, you can't see what is inside.



Task IV: DoH Spoofing (Cont.)

● Attack chain

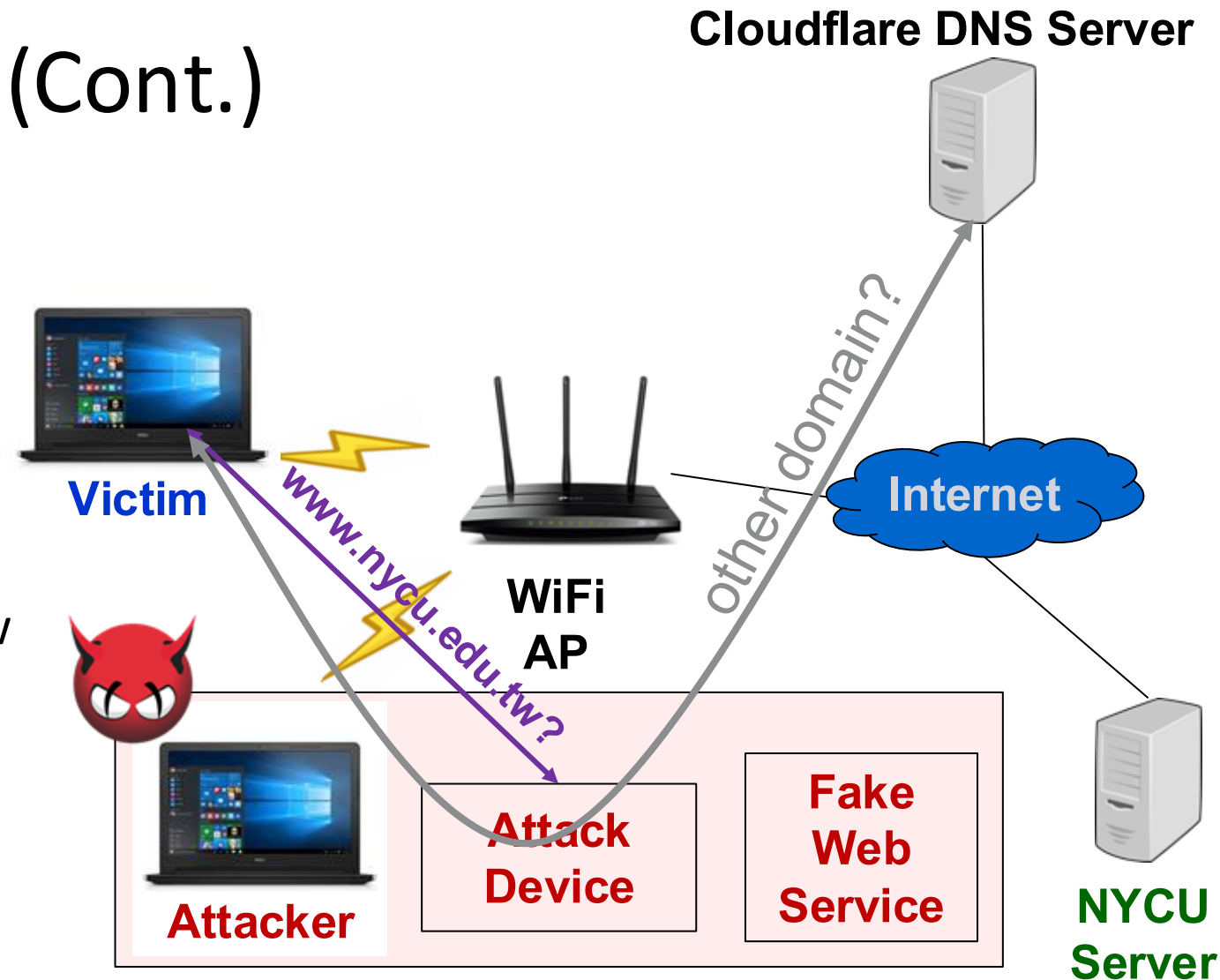
- ❑ You have to intercepts HTTPS connection instead of UDP#53.
- ❑ You act as a fake DoH server, return a spoofed DNS response pointed to the fake website.



Task IV: DoH Spoofing (Cont.)

- Handling Selective Spoofing

- ❑ Your attack device should only intercept request for www.nycu.edu.tw
- ❑ Otherwise the user would know something is wrong :P



Task IV: DoH Spoofing (Cont.)

- Prerequisites on victim VM:

- ❑ Install the provided root certificate first.
- ❑ Adjust Firefox DNS Settings to “Increased Protection”
- ❑ (Settings → Privacy&Security → scroll to section “DNS over HTTPS” → select “Increased Protection” with Cloudflare)

Enable DNS over HTTPS using:

☐ **Default Protection**
Firefox decides when to use secure DNS to protect your privacy.

☒ **Increased Protection**
You control when to use secure DNS and choose your provider.

- Use the provider you select
- Only use your default DNS resolver if there is a problem with secure DNS

Choose provider:

Cloudflare (Default)

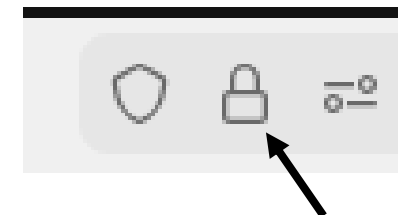
☐ **Max Protection**
Firefox will always use secure DNS. You'll see a security risk warning before we use your system DNS.



Task IV: DoH Spoofing (Cont.)

- Requirements:

- ❑ All other websites must function normally during the attack
- ❑ The browser must show no security warnings while browsing, and the padlock icon must indicate a secure connection.
- ❑ fake_server's file is also uploaded on the E3, you'll also need to sign a certificate for it.



Task IV: DoH Spoofing (Cont.)

- Hint:

- ❑ Although Firefox uses a trusted DoH provider, it still relies on the OS-level DNS resolver to obtain the initial A record of the DoH server.
- ❑ You are allow to use the given CA to sign certificates for any domain, and place it on the victim side.



Task IV: DoH Spoofing (Cont.)

- Hint:

- ❑ You may need to clear both the browser DNS cache and the system route cache to properly test the attack.
- ❑ DoH operates over HTTP/2. If your proxy does not support HTTP/2, Firefox will fail to establish a DoH connection and fall back to the system DNS resolver, bypassing your interception entirely. Make sure your implementation negotiates HTTP/2 via ALPN during the TLS handshake.



Requirements

- You need to develop/run your program in a given virtual machine
 - ❏ VM image: Please download it from [Link](#)
 - Username/password: csc2026/csc2026
 - ❏ Network setting: **NAT Network**
- Language restrictions:
 - ❏ For Task I to Task III, only C/C++ is allowed.
 - ❏ For Task IV, both Python and C/C++ are accepted.



Requirements

- Do not hardcode the network interface.
 - ☐ During the demo, the program may be run on either VMware or VirtualBox, so ensure that no fixed values are used
- You are allowed to team up. Each team has at most 2 students
 - ☐ Teams: discussions are allowed, but no collaboration
- Please only submit your source codes to E3



Important: How to Prepare Your Attack Programs?

- Must provide a **Makefile** which compiles your source codes into three executable files, named **icmp_redirect**, **pharm_attack** and **doh**(Missing: - 20%)
- Test requirements for the programs
 - ☐ Must be run in the given VM without any additional tools
 - For those using Python, use uv to manage the dependencies (You must provide toml file)
 - ☐ Must use the following parameters
 - DNS spoofing for the NYCU home page: <http://www.nycu.edu.tw> (use private window)
 - ☐ Must work for test commands



Important: How to Prepare Your Attack Programs?

- Results from the ICMP redirect attack

- ❑ Only allowed to run `./icmp_redirect <address> <interface>`
- ❑ Print out the IP/MAC addresses of **all the Wi-Fi devices/VMs** except for **Attacker**
- ❑ Select the target victim and origin gateway as the struct of fake ICMP redirect packet
- ❑ The victim should update its dynamic routing table (check by “`ip route get <address>`” on victim)

- Results from the pharming attack

- ❑ Only allowed to run `./pharm_attack` (**No manual configuration is allowed**)
- ❑ You are allowed to clean up DNS cache only one time after the first failure.
- ❑ Redirect the NYCU home page (www.nycu.edu.tw) to the phishing page
- ❑ If your attack needs firewall settings, please attach the commands in [firewall.sh](#)
- ❑ If your attack needs to modify `icmp_redirect` program, name it as `icmp_task4`.



Project Submission

- Due date: 5/6 11:55pm
 - ☐ Late submission is not allowed
- Makeup submission (75 points at most): TBA (After the final)

DUE TO LABOR DAY HOLIDAY, BOTH THE SUBMISSION DUE DATE AND DEMO HAVE BEEN PUSHED BACK BY ONE WEEK.



Project Submission

- **Submission rules** (Wrong file name or format will result in 10 points deduction)
 - ❓ Put all your files into a directory and name it using your student ID(s), with a README file describes what attack chain you implemented, and how to execute it.
 - If your team has two members, please concatenate your IDs separated by “-”
 - ❓ Zip the directory and upload the zip file to New E3
 - ❓ A sample of the zip file: 01212112-02121221.zip
 - 01212112-02121221
 - README.md
 - Makefile
 - icmp_redirect.cpp
 - pharm_attack.cpp
 - doh.cpp
 - icmp_task4.cpp
 - firewall.sh
 - 01212112-02121221
 - README.md
 - Makefile
 - icmp_redirect.cpp
 - pharm_attack.cpp
 - main.py
 - project.toml
 - icmp_task4.cpp
 - firewall.sh



Online Project Demo

- Date: 5/8
- TA will prepare your zip file and run your programs for the demo on behalf of you
- You will
 - ❑ be asked to reproduce your ICMP redirect and pharming attacks
 - ❑ be only allowed to use “make” to compile all your files, and run your attack binary programs or scripts
 - ❑ be not allowed to modify your codes or scripts
 - ❑ be asked some questions
 - ❑ be responsible to show and explain the outcome to TA



Hint 1: ICMP Redirect Attack

- ICMP redirect format

☐ Reference : [ICMP Redirect Message Format](#)

Type (8 bits)	Code (8 bits)	Checksum (16 bits)
Gateway Internet Address (32 bits)		
Internet Header + 64 bits of Original Data Datagram (Variable)		

☐ You can simply craft Internet Header + 64 bits of Original Data Datagram as an ICMP echo reply packet with a checksum set to 0xffff, and both the Identifier and Sequence Number set to 0.



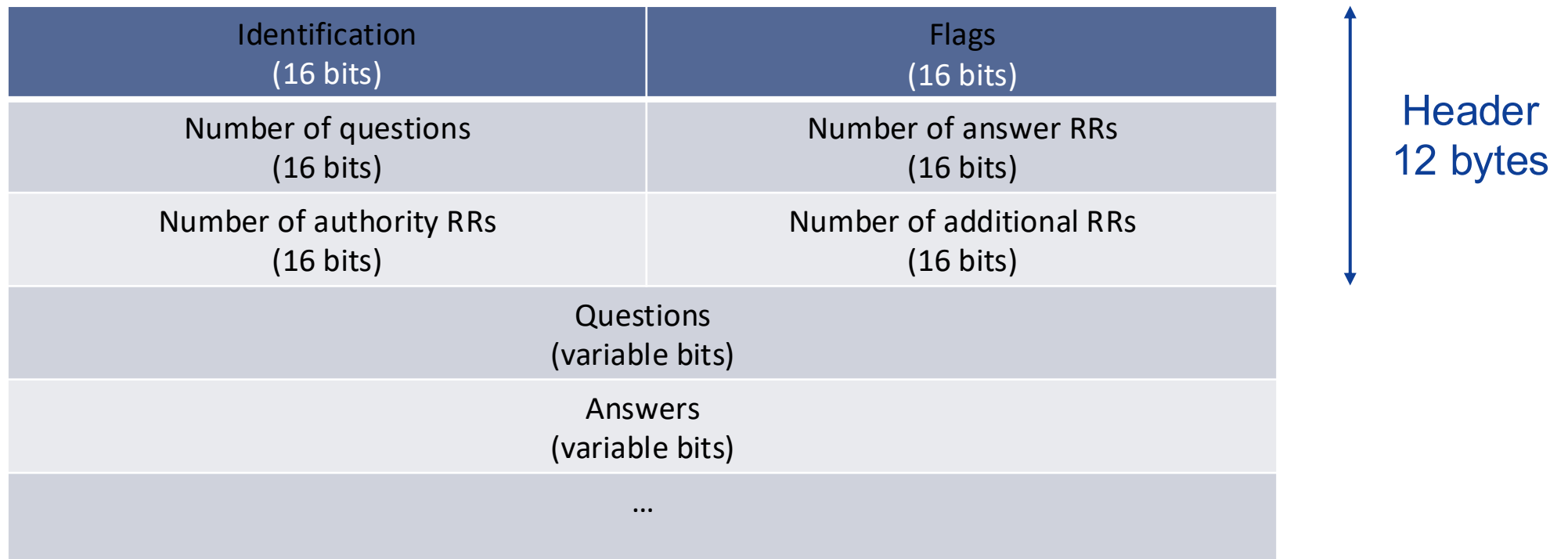
Hint 1: ICMP Redirect Attack (Cont.)

- To enable the system to accept ICMP redirects, the victim needs to configure the following network settings to 1:
 - ❑ `net.ipv4.conf.all.accept_redirects`
 - ❑ `net.ipv4.conf.default.accept_redirects`
 - ❑ `net.ipv4.conf.<interface_name>.accept_redirects`



Hint 2: DNS Spoofing

- DNS format



Hint 2: DNS Spoofing (Cont.)

- DNS packets may employ message compression
 - ▢ Reference: <https://datatracker.ietf.org/doc/html/rfc1035#section-4.1.4>



Hint 3: Netfilter queue

- Functions:

- ❑ **nfq_open():** open a nfqueue handler.
- ❑ **nfq_close():** close a nfqueue handler.
- ❑ **nfq_unbind_pf():** unbind nfqueue handler from a protocol family.
- ❑ **nfq_bind_pf():** bind a nfqueue handler to a given protocol family.
- ❑ **nfq_create_queue():** create a new queue handle and return it.
- ❑ **nfq_destroy_queue():** destroy a queue handle.
- ❑ **nfq_set_mode():** set the amount of packet data that nfqueue copies to userspace.
- ❑ **nfq_fd():** get the file descriptor associated with the nfqueue handler.
- ❑ **nfq_handle_packet():** handle a packet received from the nfqueue subsystem.
- ❑ **nfq_get_msg_packet_hdr():** return the metaheader that wraps the packet.
- ❑ **nfq_get_payload():** get payload.
- ❑ **nfq_set_verdict():** issue a verdict on a packet.

- Reference: https://netfilter.org/projects/libnetfilter_queue/doxygen/html/index.html



Hint 3: Netfilter queue(Cont.)

- `nfq_set_verdict(struct nfq_q_handle *qh, uint32_t id, uint32_t verdict, uint32_t data_len, const unsigned char *buf):`
 - ❑ **Only read the packet:** `data_len=0, *buf = nullptr`
 - ❑ **Modify packet:** `data_len=new packet length, *buf = Head of buffer that stored new packet data`



Reference 1: ICMP Redirect Attack

- This paper provides guidance on how to probe the victim in order to construct an accurate Internet Header along with the first 64 bits of the Original Data Datagram.
 - ❏ ["Man-in-the-Middle Attacks without Rogue AP: When WPA's Meet ICMP Redirects," 2023 IEEE S&P, San Francisco, CA, USA, 2023](#)
- In this project, we **don't** need to do it.



Questions?

